

PatchTST on Electricity: Reproduction, Masked SSL, and Scale Sensitivity

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PatchTST/64 reproduction

Masked SSL + FT

Scale sensitivity analysis

1. PatchTST and model pipeline

PatchTST intuition

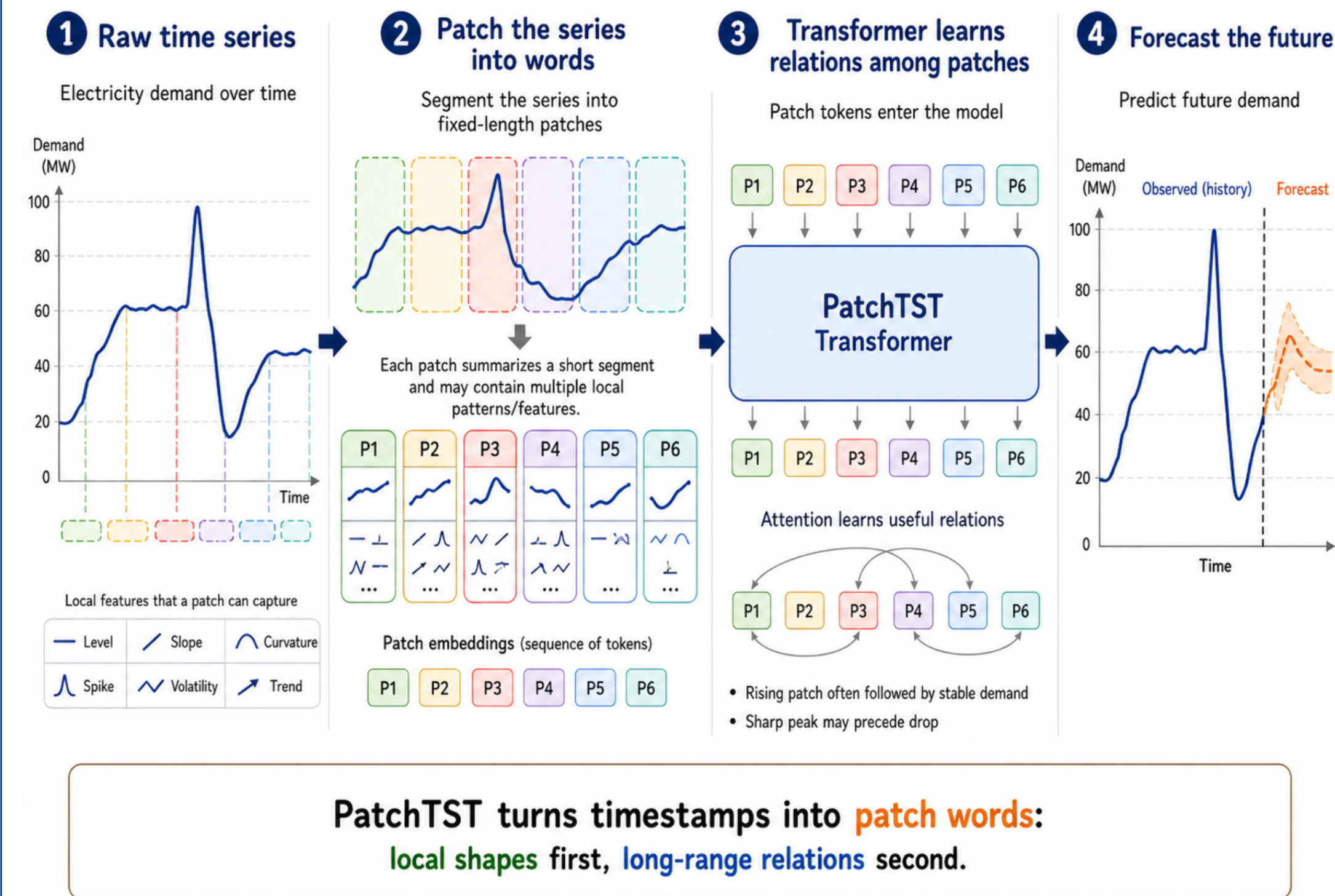


Figure 1. Cartoon of PatchTST

Paper pipeline

- **Patchifying**: instance normalisation + fixed patch length
- **Embedding**: projection + position encoding
- **Transformer**: multi-head attention
- **Linear Head**: Forecast/Mask reconstruction

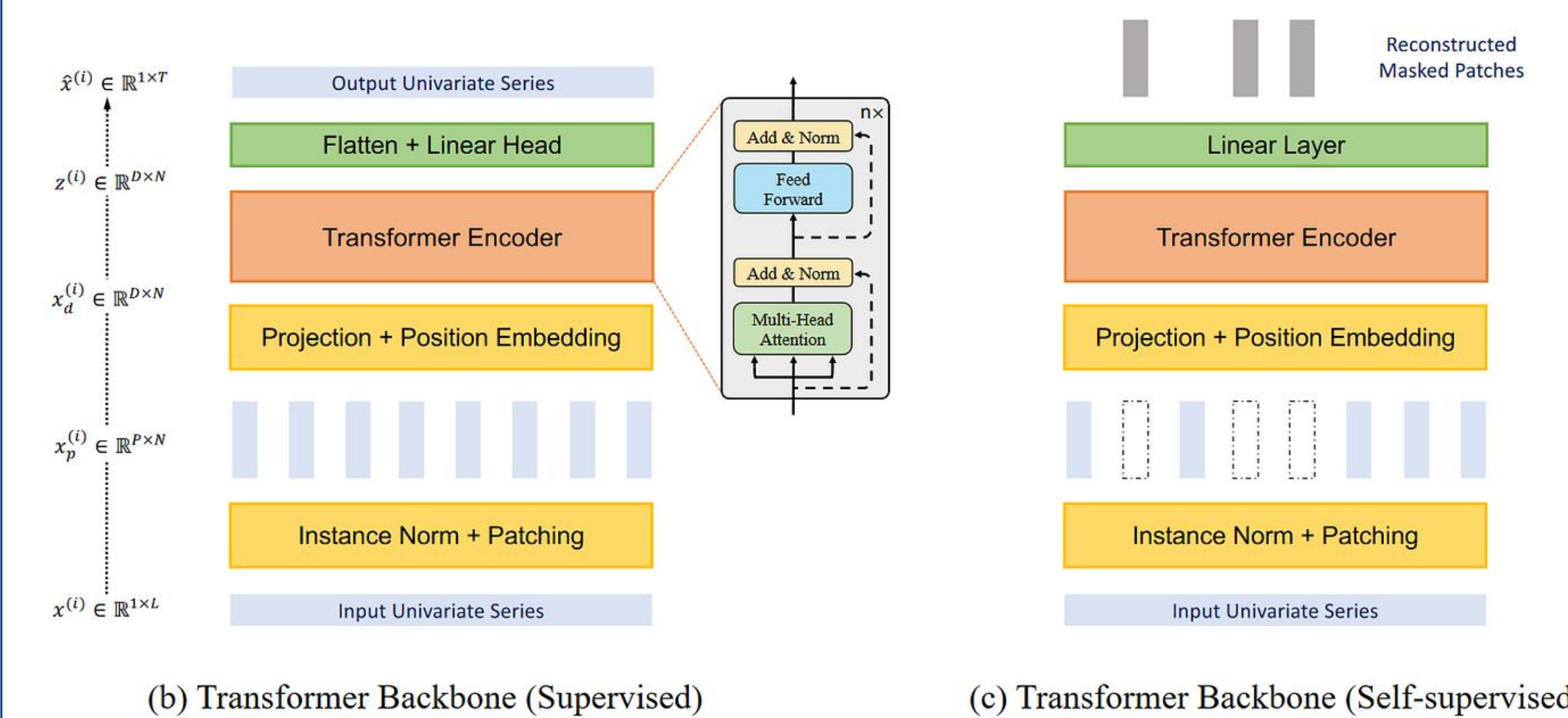
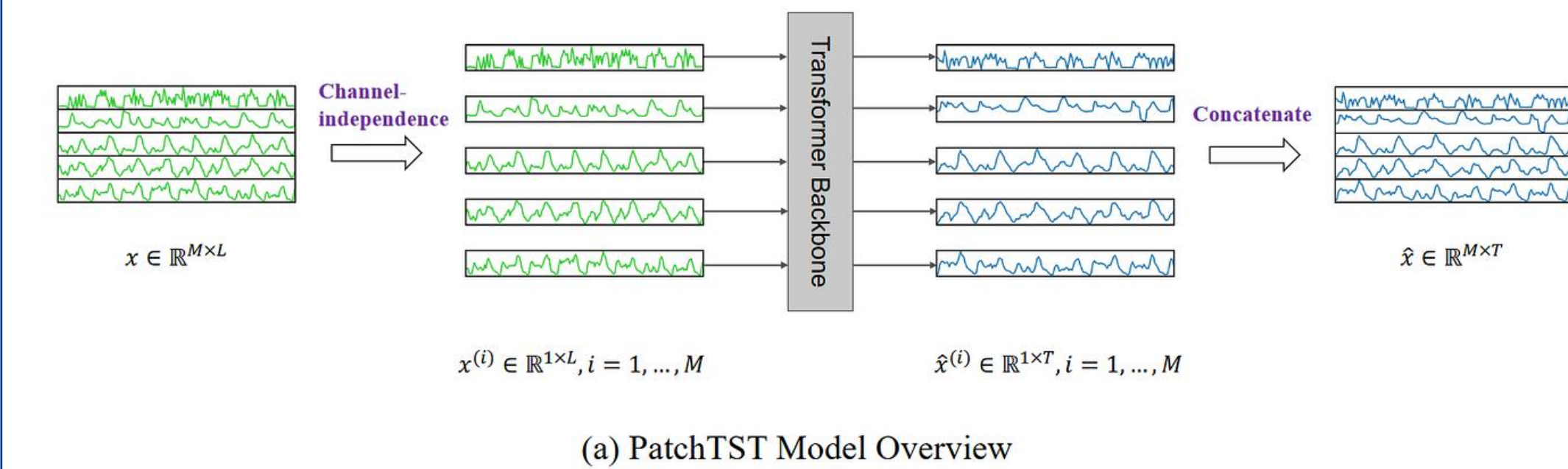


Figure 2. Original paper pipeline: channel-independent patching, Transformer

2. Forecasting results

Reproduction result

All runs are compared to the reported **PatchTST/64 paper baseline**. The metric is **test MSE**; lower is better.

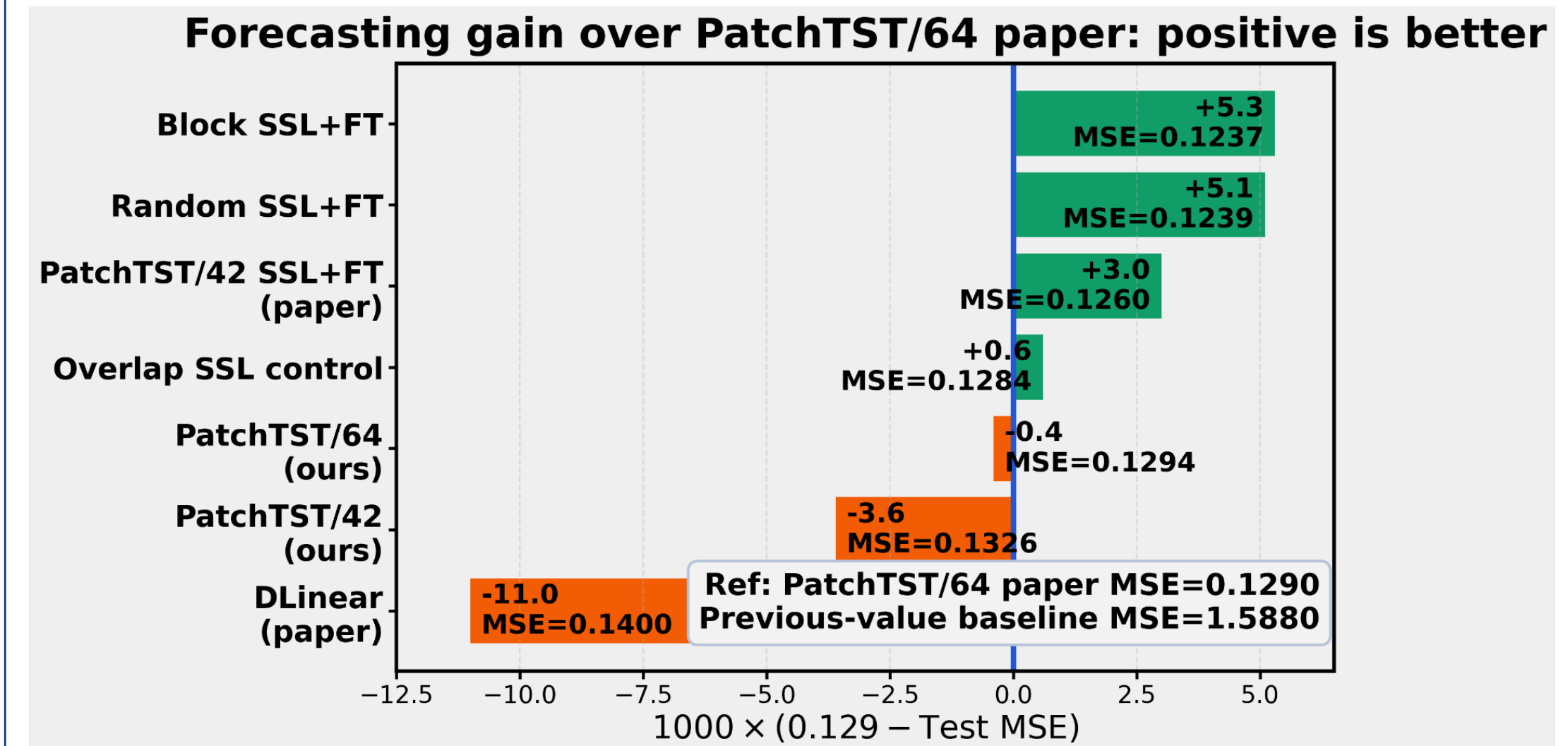


Figure 3. **Positive gain** means lower test MSE than the PatchTST/64 paper baseline.

Main result

- **PatchTST/64** nearly matches the paper: 0.1294 vs. 0.1290.
- **Random SSL+FT** improves over supervised training and the paper baseline.
- **Block SSL+FT** is best in our runs: MSE = 0.1237.
- **Overlap SSL** gives a smaller gain, so masking strategy matters.

Experiment configuration

Method	L	H	P	S	Pre	FT
Previous value	512	96	—	—	0	0
PatchTST/42	336	96	16	8	0	30
PatchTST/64	512	96	16	8	0	30
Random SSL+FT	512	96	12	12	30	15
Block SSL+FT	512	96	12	12	30	10
Overlap control	336	96	16	8	30	15

L = lookback window, H = prediction horizon, P = patch size, S = stride. Supervised runs use only **30 epochs**; SSL uses pretraining plus short fine-tuning.

Why patch words help

- **Patchify**. Reduce local noise
- **Longer lookback**. Quadratically lower computation
- **Channel-independence**. More variety of attention pattern

3. Scale sensitivity analysis

Motivation

Patch words smooth over local noise, so small scale coarse-graining may not hurt.

Scale probe: where does this patch-level robustness break?

Methodology

- Coarse-grain the input by level $\lambda = e^s$.
- Resize back to length 512; evaluate without retraining.
- Track raw-target MSE ratio against the original input.

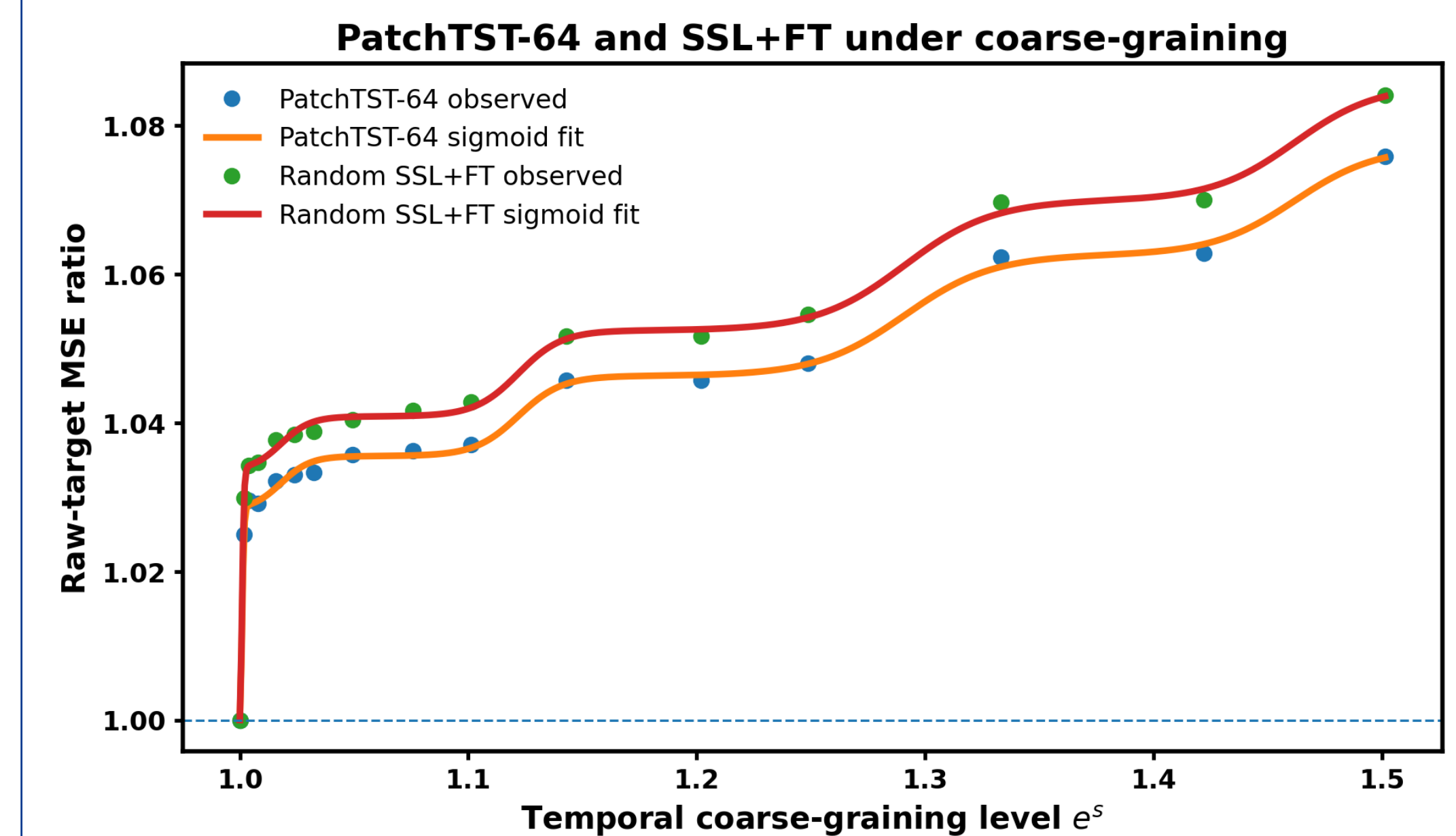


Figure 4. Higher ratio means stronger degradation after coarse-graining.

Conclusion

- Both models degrade as coarse-graining increases.
- The rise is **step-like**, not smooth.
- The representation appears **scale-structured**, not scale-invariant.

Future Directions

Probe latent patch features; test adaptive multi-scale patches; vary horizon H .

References

- PatchTST**. Nie, Nguyen, Sinthong, and Kalagnanam. "A Time Series Is Worth 64 Words: Long-term Forecasting with Transformers." ICLR 2023.
- Electricity data**. Trindade, A. (2015). *ElectricityLoadDiagrams20112014* [Dataset]. UCI Machine Learning Repository. doi:10.24432/C58C86.